RANDOM COMMUNICATION SYSTEMS BASED ON ALPHA-STABLE PROCESSES

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ABSTRACT

RANDOM COMMUNICATION SYSTEMS BASED ON ALPHA-STABLE PROCESSES

This thesis presents alpha-stable carrier based random communication systems (RCSs) as an alternate way to perform covert transmission. The first objective is to develop an optimized model of RCS which consists of a receiver that requires less computational complexity and outperforms the previously proposed receivers. Next, in order to solve the existing synchronization issue in RCSs, the general behavior of fractional lower-order covariance method in α-stable noise environments has been evaluated to establish synchronization in RCSs. An optimized range of values for the associated parameters of α-stable carrier has also been presented to optimize the proposed synchronization method.

The second objective is to establish criteria for evaluating and quantifying the security and covertness of RCSs. Therefore, the first security performance tradeoff characteristics (SPTC) have been proposed to compare the security of different RCSs. Moreover, the proposed optimized model of RCS has also been analyzed with respect to the developed security scale, i.e. SPTC. Secondly, the criterion to quantify the covertness of RCSs has also been developed to analyze the proposed RCS. Thirdly, an attack for RCS has also been proposed which highlights the potential vulnerabilities of RCSs. However, the counter-measure guidelines have been prescribed to further enhance the security of RCSs.

An inverse system approach has been adopted to propose α-stable noise driven linear time invariant system based transmitter and its corresponding inverse system based receiver as a third objective. It can be considered as the most secure model for α-stable noise carrier based RCS till now.

ÖZET

ALFA-KARARLI SÜREÜÇLER TABANLI RASSAL HABERLEŞME SİSTEMİ

Bu tez güvenli kablosuz haberleşme sistemlerine alternatif olacak alfa kararlı (α-kararlı)taşıyıcı tabanlı rassal haberleşme (RCS) sistemi sunmaktadır.Bu nedenle ilk hedef daha önce sunulmuş alıcılara göre bit hata oranını (BER) daha iyileştirecek daha az hesapsal karmaşıklık gerektiren etkin bir alıcı içeren optimal bir (RCS) geliştirmektir. Daha sonra RCS sistemlerinde daha önce ele alınmamış senkronizasyon problemini çözümlemek için alfa-kararlı ortamlar için geliştirilmiş. Kesirli düşük-mertebeli kovaryans metodu literatürdeki ilk senkonize RCS (SRCS)'i sunmak üzere kullanılmıştır.Daha sonra bu sunulan SRCS'nin BER performansını arttırmak amacıyla alfa-kararlı gürültü taşıyıcının parametrelerinin en uygun değerleri hesaplanmıştır.

İkinci hedef de RCS'nin güvenilirliğini ve gizliliğini nitelendirip değerlendirecek kriterler geliştirmek olmuştur. Bu nedenle ilk önce literatürde varolan diğer RCS modellerinin güvenlikleriyle BER performanslarını karşılaştırmak için Güvenlik Performans Tercih Karakteristikleri (STPC) sunulmuştur. Sunulan optimize edilmiş RCS bu geliştirilen güvenlik ölçüsü (STPC)ne göre analiz edilmiş ve literatürde varolan tasarımlara göre daha güvenli olduğu gösterilmiştir. Ikinci aşama olarak RCS'nin güvenilirliğini nitelendirmek için bir kriter geliştirilmiş ve sunulan optimize edilmiş RCS "istenmeyen dinleyici"açısından incelenmiştir. Üçüncü aşamada RCS'nin olası kırılganlığını sınamak için istenmeyen dinleyicinin RCS'ye saldırıları tasarlanmıştır. Bununla birlikte RCS'nin güvenirliliğini arttırmak için gizlilik aralığı ve karşı tedbir ölçüleri için bir rehber belirlenmiştir.

Karşı tedbir ölçüsü rehberi geliştirdikten sonra tezde üçüncü aşama olarak α-kararlı gürültü ile sürülmüş doğrusal zamanla değişmeyen sisteme dayali verici ile bu doğrusal vericinin tersini alıcı olarak tasarlayan "Ters Sistem" tabanlı bir yeni RCS sunulmuştur. Bu Ters Sistem tabanlı RCS de saldırılara karşı daha az kırılgan yapısı nedeniyle varolan alfa-kararlı gürültü taşıyıcı tabanlı tüm RCS'lere göre daha güvenilir olarak düşünülmektedir.Tez içinde her aşamada yapılan benzetimlerden elde edilen sonuçlar yapılan çalışmaların kullanılabilme gücünü göstermek için sunulmuştur.

To my wife Rida and my sons Abyan and Raaed.....

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α-stable	Alpha-stable
ACR	Achieved Confidence Ratio
ARMA	Autoregressive/Moving Average
BER	Bit Error Rate
CDMA	Code Division Multiple Access
CR	Confidence Ratio
CM	Chaotic Masking
CSK	Chaotic Shift Keying
DCSK	Differential Chaotic Shift Keying
DDCSK	Differentially Differential Chaotic Shift Keying
DCSK-CS	Differential Chaotic Shift Keying (Code Shifted)
DCSK-HE	Differential Chaotic Shift Keying (High Efficiency)
DCSK-I	Differential Chaotic Shift Keying (Improved)
DCSK-MC	Differential Chaotic Shift Keying (Multicarrier)
DCSK-PMA	Differential Chaotic Shift Keying (Permutation based)
DCSK-PS	Differential Chaotic Shift Keying (Phase seperated)
DCSK-RM	Differential Chaotic Shift Keying (Reference Modulated)
DCSK-WC	Differential Chaotic Shift Keying (Walsh Code based)
DCSK-UWB	Differential Chaotic Shift Keying (Ultra Wideband)
DS-SS	Direct Sequence Spread Spectrum Communication
FH-S	Frequency Hoping Spread Spectrum Communication
FLOC	Fractional Lower-Order Covariance
FLOACB	Fractional Lower-Order Auto-Covariance Block
FLOCC	Fractional Lower-Order Covariance based Correlators
H-SS	Hybrid Spread Spectrum
LTI	Linear Time Invariant
MEVM	Modified Extreme Value Method
MSNR	Mixed Signal to Noise Ratio
PAS	Pilot Assisted Synchronization
PDF	Probability Density Function
RCS	Random Communication System

RCSR	Random Carrier Signal Recognizer
RCSRI	Random Carrier Signal Recognition Indicator
SB	Synchronization Block
SCS	Stochastic Communication System
SG	Signal Generator
SKαS-NSK	Skewed Alpha-stable Noise Shift Keying
SPTC	Security Performance Tradeoff Characteristics
SRCS	Synchronized Random Communication System
TCB	Threshold Control Block
TD	Threshold Detector
TH-SS	Time Hopping Spread Spectrum Communication

CHAPTER 1

INTRODUCTION

1.1. Overview

Security of communication systems has always remained a key topic among scientists, researchers and engineers associated with the field of telecommunications; where their huge emphasis has always been given to establish secure wireless communication systems. In the last few decades, physical layer secured wireless communication systems has gained much attention all over the globe which reflects the potential of the idea to ensure covert transmission. Due to the rapid growth and demand of wireless communication systems in the last decades, increment in inter and cross disciplinary approaches between wireless, vehicular, robotics, wearable electronics and nano devices, for building future communication systems are also on the rise. Similarly, the never ending amount of wireless communication devices, to explore, apply and create digital dividends to build a fully connected world in future, are also escalating the security concerns. Therefore, growing interest in enhancing the physical layer security of wireless communications systems indicates that this field has the capability to significantly eliminate the security problems for next-generation wireless communication systems.

Many conventional approaches have been used in the past to increase the security of classic sinusoidal carrier based wireless communication systems; spread spectrum (SS) communication is one of those as it works by maneuvering the parameters of conventional sinusoidal carrier Abdallah et al. (1991), Peterson et al. (1995) and Salehi, and Proakis (2007). SS communication based on Pseudo Random (PN) noise became popular due to their increased physical layer security as well as better BER performance. However, different algorithms, e.g. linear regression attack and exploiting the drawback of short linear complexity of SS signals in Tsatsanis and Giannakis (1997), Burel and Bouder (2000) and Burel, G. (2000), had been proposed later on to overcome the security provided by SS signals. This shortcoming had to be overcome by utilizing different types of noise like signals. This was the period when chaotic signals, i.e. non-periodic noise signals consisting of infinite number of states, were utilized to remove the weak holes in the security provided by SS systems. Among the unconventional approaches, chaotic communications became popular, after the discovery of the first method to synchronize chaos in 1991 by Pecora and Carroll (1990), as they used chaotic noise signals as carrier to encrypt the information content in the transmitted signals Oppenheim et al. (1992) and Dedieu et al. (1993) .However, chaotic communication systems besides having serious security vulnerabilities, proposed till now, are only able to hide the information content in the channel, but the eavesdroppers are aware of the existence of communication Sobhy and Shehata (2001). Due to the fact that the security of chaotic communication systems is derived in this way, many blind and semi-blind signal recognition techniques based on geometric forecasting approach, synchronizing circuits, auto-correlation, generalized synchronization, power analysis and signal filtering had been proposed to completely decode the information content hidden in the noise like chaotic carrier signals Short (1994), Short, K.M. (1996), Sobhy and Shehata (2001), Alvarez et al. (2004), Alvarez et al. (2004b) and Chien and Liao (2005),. The drawbacks in noise like chaotic carrier based communication systems established the need to introduce some actual noise carrier based communication systems.

Since, using noise signals as random carrier can not only hide the communication content, but it also makes the eavesdroppers unaware of its existence, hence, an eavesdropper has no idea whether a meaningful message is being sent or not Gu and Cao (2012). Therefore, efforts to use noise as a carrier to establish more secure spread spectrum communication system started in 1950's Basore (1952). However, a complete stochastic communication system based on stochastic process shift keying (SPSS) was first introduced by Salberg and Hanssen (1999). In their system, two different Autoregressive/Moving Average (ARMA) processes were used to send binary messages.

The α-stable noise, being the generalized version of the Gaussian noise, has the potential to be utilized as a random carrier as well as has the capability to be invisible in AWGN channel; where the invisibility is as severe as close the parameters of the transmitted α-stable random noise carrier to the parameters of the Gaussian noise present in the channel. However, detection and parameter estimation of α-stable random noise signals is considered to be a challenging problem in signal processing due to common characteristic properties of all heavy-tailed stable distributions which are i) the nonexistence of finite second or higher-order moments; ii) relatively high probabilities of large deviations from the median Nikias and Shao (1995). Moreover, α-stable noise signal waveform contains no repetitions or periodicities, so the pulse length is also hidden which makes it a natural candidate for secure communication. Therefore, many signal processing methods have been proposed in the past to estimate the related parameters of α-stable noise signals Ma and Nikias (1996), Tsihrintzis and Nikias (1996), Kuruoğlu (2001) and Kannan and Ravishanker (2007) or to model and estimate the number of α-stable distributions in mixture distributions Casarin (2004) and Diego et al. (2009) but utilizing them to perform covert transmission requires a priori knowledge of the utilized carrier signals. Therefore, the idea to use α-stable noise as random carrier to covertly convey the binary information provides physical layer security during transmission; which reduces the risk of eavesdropping.

This fact gave birth to symmetric and skewed α-stable random noise carrier based binary communication systems, i.e. random communication systems (RCSs) by Cek and Savaci (2009) and Cek (2015). Different receiver designs, signal detection and estimation schemes have also been utilized to increase the efficiency of RCSs; where it has also been analyzed that RCSs have the capability to perform covert transmission in fading channels as well by Cek (2015b), Xu et al. (2016) and Xu et al. (2017). However in this thesis, we have introduced an optimized model of RCS followed by the first criterion to quantify the covertness of α-stable noise based communication systems Ahmed and Savaci (2017) and Ahmed and Savaci (2017b). All studies related to RCSs have already assumed perfect synchronization; where no exact method was presented until we introduced the first method to synchronize RCSs followed by a criterion to optimize SRCSs Ahmed and Savaci (2018) and Ahmed and Savaci (2018b) which are also covered in the thesis as one of the key objectives. The security of the proposed models of RCSs is derived from the facts that i) even though α-stable mixtures can be separated by the Bayesian techniques presented by Ma and Nikias (1996) and Tsihrintzis and Nikias (1996), the pulse length cannot be estimated therefore α-stable random noise signals are undetectable or invisible to an eavesdropper; ii) the pulse length, i.e. duration of the carrier signal holding single binary information bit, needed for decoding is also hidden. These facts make it extremely difficult for an eavesdropper to blindly recognize and estimate the related parameter of α-stable carrier signals to retrieve the transmitted binary information.

In RCSs, decoding the received signal is considered extremely difficult without knowing the pulse length, i.e. duration of a single binary information bit. However, if the exact pulse length can be found then it would be possible for an eavesdropper to retrieve the binary messages hidden in the transmitted α-stable noise signals by one of the estimation methods given by Kuruoğlu (2001). Previously, there was not any attack, algorithm available to crack the hidden pulse length involved in conveying the binary information. In this thesis, the first attack, i.e. blind detection, recognition and data extraction method, for α-stable carrier signals has been proposed and analyzed from the perspective of an eavesdropper. The proposed attack first detects the possible presence of α-stable random carrier signals and then recognizes the associated impulsiveness and skewness parameters, exploited by the transmitter and the intended receiver, to extract covertly conveyed binary information. However, the covert range has also been prescribed which can be adopted to perform secure transmission by RCSs.

In order to assure true level of covertness by RCSs, an inverse system approach has also been adopted to design RCS as the main objective of the thesis which is prone to the proposed attack. Since, the newly developed RCS includes the mth-order linear time invariant (LTI) dynamical system with Sk-αSNSK signal generator (SG) at the transmitting end 'Alice' while the intended receiver 'Bob' uses the inverse LTI dynamical system of the transmitter with the modified extreme value method (MEVM) based estimator, hence, the parameters needed to decode the information carrying α-stable input has been increased from single parameter, i.e. pulse length, to the state parameters of the utilized LTI dynamical system as well. Hence, estimation of skewness parameter by an eavesdropper, without using the inverse system, will not reveal the true binary messages while the intended receiver truly decodes the binary messages. The improvement in security is shown by comparing the bit error rate performances of the intended receiver and an eavesdropper.

Therefore, in this thesis, the authors have tried to contribute a lot towards this field of RCSs which is still in its infancy. However, due to very small number of researchers working in this area, there is still a lot of work which can be done. These ideas or suggestion have been given in the end of this thesis. Before going further, the core of this thesis, i.e. α-stable noise or distribution, and its associated properties and parameter have been explained in detailed below.

1.2. Alpha-Stable Noise Distribution

Alpha-stable (α-stable) processes, especially Gaussian processes, have long being used as a useful tool to model various stochastic processes. Efforts to develop the theory of single variable stable distributions started in the 1920s and 1930s by Paul Levy and Aleksander Yakovlevich Khinchine. Classic form of stable distribution is covered in Gnedenko and Kolmogorov (1954) and Feller (1971), where it has been updated in monograph of Zolotarev (1986). A review of the state of the art on stable processes from a statistical point of view is provided by a collection of papers edited by Cambanis et al. (1991). The definitions, mathematical description and recent application of α-stable distribution are reviewed below.

Definition: As defined in Samorodnitsky and Taqqu (1994), if $\triangleq$ denotes equality in distribution and the relation

$$X_1 + X_2 + \ldots\ldots + X_n \triangleq C_n X + D_n \tag{1}$$

holds for a positive constant C_n and a real number D_n given that X_1, X_2, ……, X_n are independent and identically distributed random variables then X is said to follow and α-stable distribution.

1.2.1. Mathematical Description

The characteristic function of α-stable Levy noise $X \sim S_\alpha(\beta, \gamma, \mu)$ having α-stable distribution is expressed in Samorodnitsky and Taqqu (1994) as

$$\phi(\theta) = \begin{cases} \exp\{j\mu\theta - \gamma^{\alpha}\,|\theta|^{\alpha}(1 - j\beta sign(\theta) tan\left(\frac{\alpha\pi}{2}\right))\} \ if\ \alpha \neq 1 \\ \exp\{j\mu\theta - \gamma\,|\theta|(1 + j\beta\frac{2}{\pi} sign(\theta) ln\left(\frac{\alpha\pi}{2}\right))\} \ if\ \alpha = 1 \end{cases} \tag{2}$$

where

$$sign(x) = \begin{cases} -1 & if\ x < 1, \\ 0 & if\ x = 0, \\ 1 & if\ x > 1. \end{cases} \tag{3}$$

and the four related parameters are defined in the respective ranges as : the characteristic exponent α ($0 < \alpha \leq 2$), the skewness parameter β($-1 \leq \beta \leq 1$), the dispersion parameter γ ($\gamma \geq 0$) and the location parameter $\mu \in R$.

Remark 1: Gaussian, Cauchy and Levy distributions are the special cases of α-stable distributions defined as X ~ $S_{\alpha=2}$ $(\beta=0, \gamma, \mu)$, X ~ $S_{\alpha=1}$ $(\beta=0, \gamma, \mu)$ and X ~ $S_{\alpha=0.5}$ $(\beta = \mp 1, \gamma, \mu)$, respectively.

Note: The second or higher-order moments do not exist for $\alpha < 2$, moreover, the first order moment do not exist as well for $\alpha \leq 1$. The α-stable signals can be generated by utilizing the characteristic function of α-stable noise 'X', i.e. X ~ S_α (β, γ, μ), given in Janicki and Weron (1994).

Properties: Samorodnitsky and Taqqu (1994), Property 1.2.2 and 1.2.3,

Let X ~ S_α (β, γ, μ) and let h be a non-zero real constant. Then

$$h + X \sim S_\alpha (\beta, \gamma, h+\mu) \tag{4}$$

and

$$h\, X \sim S_\alpha (sign\{h\}.\beta, |h|\, \gamma, h\mu) \tag{5}$$

Types: The frequent occurrence of extreme values in the observed data of many actual physical phenomena is non-Gaussian, hence; they are considered as a random process and are modeled by α-stable distribution. However, symmetric α-stable (SαS) process indicates that positive and negative outcomes in the observed data are equally likely. However, Skewed α-stable (SkαS) process indicates that the observed data is biased towards either positive or negative outcomes. Therefore, α-stable distribution can be categorized in to two main types depending on the type of the parameter used to shape up its probability density function (PDF) as

1. Distribution dependent on α parameter is known SαS distribution which is shown in Figure 1.1.
2. Distribution dependent on β parameter is known SkαS distribution which is shown in Figure 1.2.

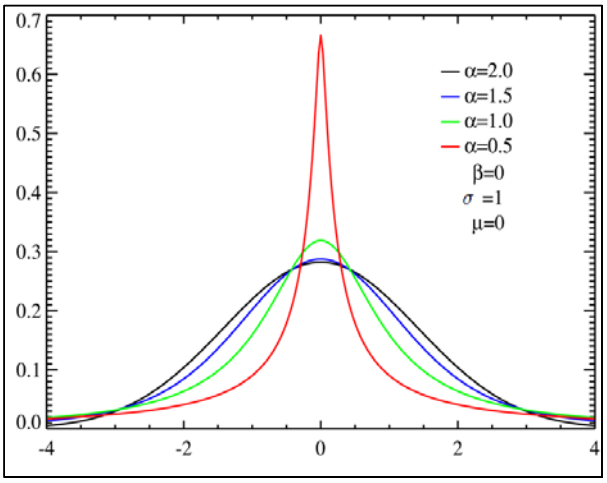

Figure 1.1. Symmetric α-stable distributions with unit scale factor

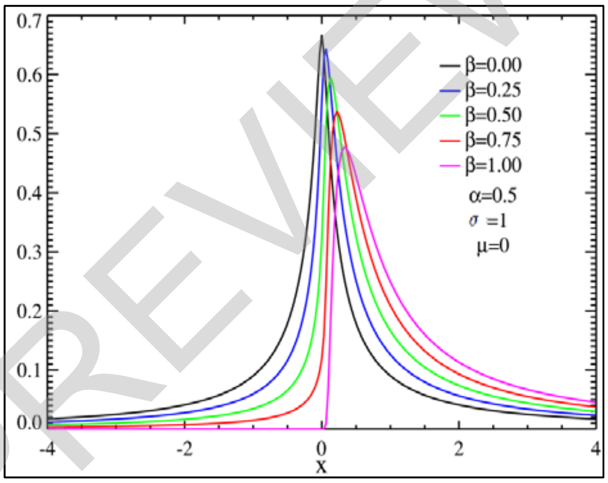

Figure 1.2. Skewed centered stable distributions with unit scale factor

1.2.2. Applications of Alpha-stable noise

As α-stable distribution is expected from superposition in natural processes, therefore it has been used in many hot areas for investigations. There are plenty of sources that follow or be modeled by α-stable distribution which includes seismic activity, ocean wave variability, lightning in the atmosphere, climatology and weather,